

LAB Assignment

(Studio 3T)

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CSF 017

COLLECTION 1: **flights**

Schema

```
{  
  flight_id: Number,  
  airline: String,  
  source: String,  
  destination: String,  
  price: Number,  
  duration: Number  
}
```

Table

flight_id	airline	source	destination	price	duration
1	Indigo	Delhi	Mumbai	5000	2
2	AirIndia	Kolkata	Chennai	6500	3
3	SpiceJet	Mumbai	Bangalore	4000	1.5
4	Vistara	Delhi	Chennai	7000	3
5	GoFirst	Mumbai	Delhi	4800	2
6	Indigo	Bangalore	Kolkata	5500	2.5

COLLECTION 2: **bookings**

Schema

```
{  
  booking_id: Number,  
  passenger_name: String,  
  flight_id: Number,  
  city: String,  
  seats: Number,  
}
```

```
total_fare: Number
}
```

Table

booking_id	passenger_name	flight_id	city	seats	total_fare
101	Aarav	1	Delhi	1	5000
102	Meera	2	Kolkata	2	13000
103	Rohan	3	Mumbai	1	4000
104	Sneha	4	Delhi	3	21000
105	Kunal	5	Mumbai	2	9600
106	Priya	6	Bangalore	1	5500

QUESTIONS

Q1. Display all flights.

```
use airlineDBCSF017
db.flightsCSF017.insertMany([
{ flight_id: 1, airline: "Indigo", source: "Delhi", destination: "Mumbai", price: 5000, duration: 1.5},
{ flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Chennai", price: 6500, duration: 2},
{ flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Bangalore", price: 4000, duration: 1.5},
{ flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai", price: 7000, duration: 3},
{ flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi", price: 4800, duration: 1.5},
{ flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolkata", price: 5500, duration: 1.5}
])
db.bookingsCSF017.insertMany([
{ booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", seats: 1, total_fare: 5000},
{ booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata", seats: 2, total_fare: 13000},
{ booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai", seats: 1, total_fare: 4000},
{ booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", seats: 3, total_fare: 21000},
{ booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai", seats: 2, total_fare: 9600},
{ booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore", seats: 1, total_fare: 5500}
])
db.flightsCSF017.find()
```

```
{
  "price": NumberInt(5000),
  "duration": NumberInt(2)
}
{
  "_id": ObjectId("69d00ea3f6e23af6e444152f"),
  "flight_id": NumberInt(2),
  "airline": "AirIndia",
  "source": "Kolkata",
  "destination": "Chennai",
  "price": NumberInt(6500),
  "duration": NumberInt(3)
}
{
  "_id": ObjectId("69d00ea3f6e23af6e4441530"),
  "flight_id": NumberInt(3),
  "airline": "SpiceJet",
  "source": "Mumbai",
  "destination": "Bangalore",
  "price": NumberInt(4000),
  "duration": 1.5
}
{
  "_id": ObjectId("69d00ea3f6e23af6e4441531"),
  "flight_id": NumberInt(4),
  "airline": "Vistara",
  "source": "Delhi",
  "destination": "Chennai",
  "price": NumberInt(7000),
  "duration": NumberInt(3)
}
```

Q2. Display passenger_name and city from bookings.

```
use airlineDBCSF017
db.flightsCSF017.insertMany([
  { flight_id: 1, airline: "Indigo", source: "Delhi", destination: "Mumbai", price: 5000, duration: 2 },
  { flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Chennai", price: 6500, duration: 3 },
  { flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Bangalore", price: 4000, duration: 1 },
  { flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai", price: 7000, duration: 3 },
  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi", price: 4800, duration: 1 },
  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolkata", price: 5500, duration: 2 }
])
db.bookingsCSF017.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", seats: 1, total_fare: 5000 },
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata", seats: 2, total_fare: 13000 },
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai", seats: 1, total_fare: 4000 },
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", seats: 3, total_fare: 21000 },
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai", seats: 2, total_fare: 9600 },
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore", seats: 1, total_fare: 5500 }
])
db.flightsCSF017.find()
db.bookingsCSF017.find(
  {},
  { passenger_name: 1, city: 1, _id: 0 }
)
```

```
1 {
  "passenger_name": "Aarav",
  "city": "Delhi"
}
2 {
  "passenger_name": "Meera",
  "city": "Kolkata"
}
3 {
  "passenger_name": "Rohan",
  "city": "Mumbai"
}
4 {
  "passenger_name": "Sneha",
  "city": "Delhi"
}
5 {
  "passenger_name": "Kunal",
  "city": "Mumbai"
}
6 {
  "passenger_name": "Priya",
  "city": "Bangalore"
}
7 {
  "passenger_name": "Aarav",
  "city": "Delhi"
}
8 {
  "passenger_name": "Meera"
```

Q3. Find flights where source is Delhi.

```
flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Bangalore", price: 4000, duration: 1
flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai", price: 7000, duration: 3
flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi", price: 4800, duration: 1
flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolkata", price: 5500, duration: 2
})
db.bookingsCSF017.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", seats: 1, total_fare: 5000 },
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata", seats: 2, total_fare: 13000 },
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai", seats: 1, total_fare: 4000 },
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", seats: 3, total_fare: 21000 },
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai", seats: 2, total_fare: 9600 },
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore", seats: 1, total_fare: 5500 }
])
db.flightsCSF017.find()
db.bookingsCSF017.find(
  {},
  { passenger_name: 1, city: 1, _id: 0 }
)
db.flightsCSF017.find({ source: "Delhi" })
```

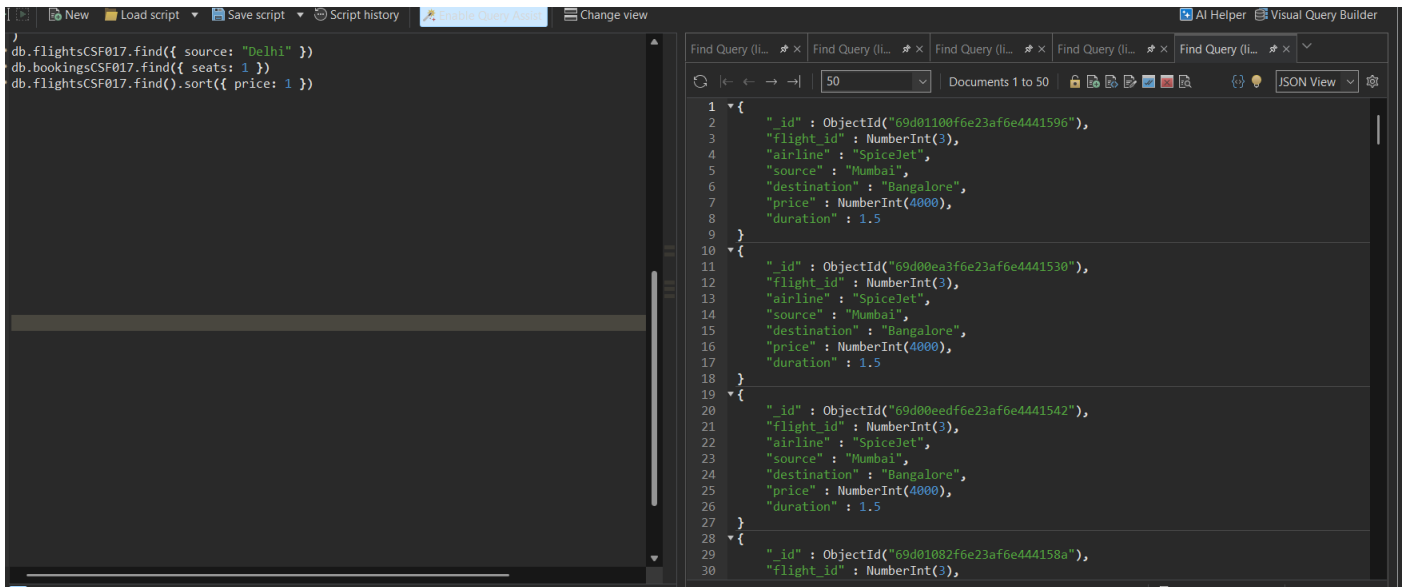
```
1 {
  "id": ObjectId("69d00ea3f6e23af6e444152e"),
  "flight_id": NumberInt(1),
  "airline": "Indigo",
  "source": "Delhi",
  "destination": "Mumbai",
  "price": NumberInt(5000),
  "duration": NumberInt(2)
}
2 {
  "id": ObjectId("69d00ea3f6e23af6e4441531"),
  "flight_id": NumberInt(4),
  "airline": "Vistara",
  "source": "Delhi",
  "destination": "Chennai",
  "price": NumberInt(7000),
  "duration": NumberInt(3)
}
3 {
  "id": ObjectId("69d00ec8f6e23af6e4441534"),
  "flight_id": NumberInt(1),
  "airline": "Indigo",
  "source": "Delhi",
  "destination": "Mumbai",
  "price": NumberInt(5000),
  "duration": NumberInt(2)
}
4 {
  "id": ObjectId("69d00ec8f6e23af6e4441537"),
  "flight_id": NumberInt(4),
```

Q4. Find bookings where seats = 1.

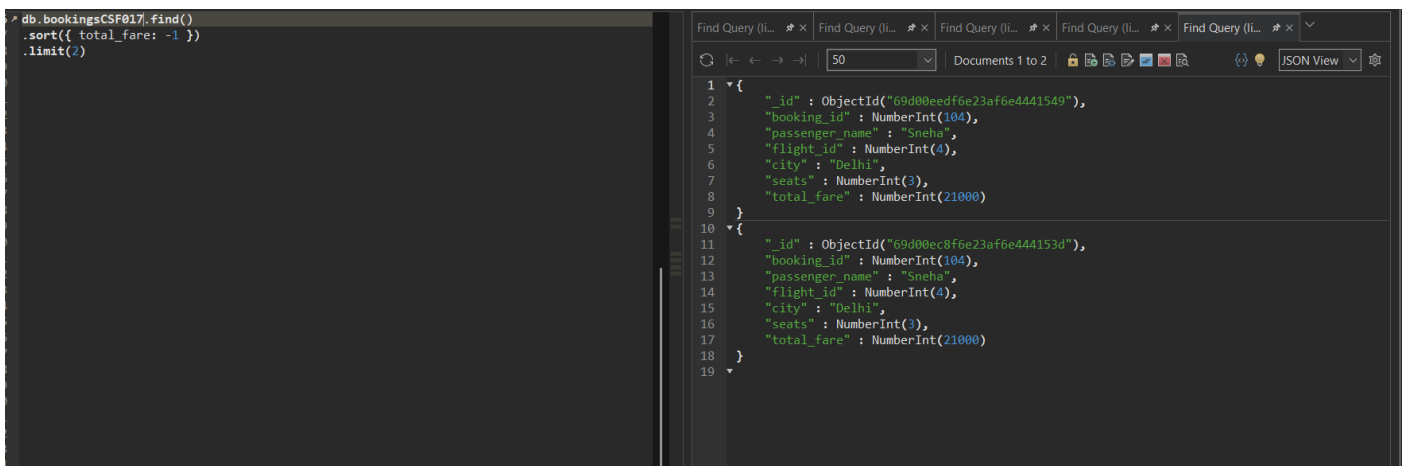
```
db.bookingsCSF017.find({ seats: 1 })
```

```
1 {
  "id": ObjectId("69d00ec8f6e23af6e444153a"),
  "booking_id": NumberInt(101),
  "passenger_name": "Aarav",
  "flight_id": NumberInt(1),
  "city": "Delhi",
  "seats": NumberInt(1),
  "total_fare": NumberInt(5000)
}
2 {
  "id": ObjectId("69d00ec8f6e23af6e444153c"),
  "booking_id": NumberInt(103),
  "passenger_name": "Rohan",
  "flight_id": NumberInt(3),
  "city": "Mumbai",
  "seats": NumberInt(1),
  "total_fare": NumberInt(4000)
}
3 {
  "id": ObjectId("69d00ec8f6e23af6e444153f"),
  "booking_id": NumberInt(106),
  "passenger_name": "Priya",
  "flight_id": NumberInt(6),
  "city": "Bangalore",
  "seats": NumberInt(1),
  "total_fare": NumberInt(5500)
}
4 {
  "id": ObjectId("69d00eedf6e23af6e4441546"),
  "booking_id": NumberInt(101),
```

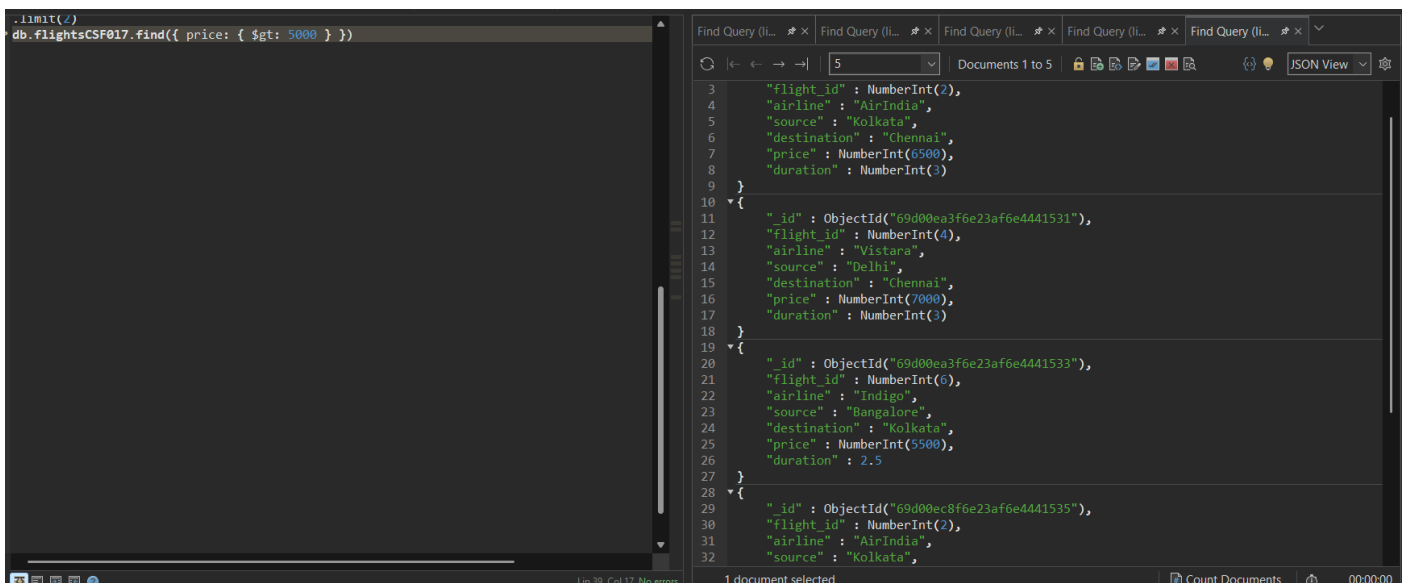
Q5. Sort flights by price in ascending order.



Q6. Display top 2 bookings with highest total_fare.



Q7. Find flights where price is greater than 5000.



Q8. Display flights where destination is not Chennai.

```
db.flightsCSF017.find({ destination: { $ne: "Chennai" } })
```

```
{
  "_id": ObjectId("69d00ea3f6e23af6e444152e"),
  "flight_id": NumberInt(1),
  "airline": "Indigo",
  "source": "Delhi",
  "destination": "Mumbai",
  "price": NumberInt(5000),
  "duration": NumberInt(2)
}
{
  "_id": ObjectId("69d00ea3f6e23af6e4441530"),
  "flight_id": NumberInt(3),
  "airline": "SpiceJet",
  "source": "Mumbai",
  "destination": "Bangalore",
  "price": NumberInt(4000),
  "duration": 1.5
}
{
  "_id": ObjectId("69d00ea3f6e23af6e4441532"),
  "flight_id": NumberInt(5),
  "airline": "GoFirst",
  "source": "Mumbai",
  "destination": "Delhi",
  "price": NumberInt(4800),
  "duration": NumberInt(2)
}
{
  "_id": ObjectId("69d00ea3f6e23af6e4441533"),
  "flight_id": NumberInt(6),
  "airline": "Indigo",
  "source": "Delhi",
  "destination": "Mumbai",
  "price": NumberInt(5000),
  "duration": NumberInt(2)
}
```

Q9. Find bookings where seats are between 1 and 2.

```
db.bookingsCSF017.find({
  seats: { $gte: 1, $lte: 2 }
})
```

```
{
  "_id": ObjectId("69d00ec8f6e23af6e444153a"),
  "booking_id": NumberInt(101),
  "passenger_name": "Aarav",
  "flight_id": NumberInt(1),
  "city": "Delhi",
  "seats": NumberInt(1),
  "total_fare": NumberInt(5000)
}
{
  "_id": ObjectId("69d00ec8f6e23af6e444153b"),
  "booking_id": NumberInt(102),
  "passenger_name": "Meera",
  "flight_id": NumberInt(2),
  "city": "Kolkata",
  "seats": NumberInt(2),
  "total_fare": NumberInt(13000)
}
{
  "_id": ObjectId("69d00ec8f6e23af6e444153c"),
  "booking_id": NumberInt(103),
  "passenger_name": "Rohan",
  "flight_id": NumberInt(3),
  "city": "Mumbai",
  "seats": NumberInt(1),
  "total_fare": NumberInt(4000)
}
{
  "_id": ObjectId("69d00ec8f6e23af6e444153e"),
  "booking_id": NumberInt(105),
  "passenger_name": "Aarav",
  "flight_id": NumberInt(1),
  "city": "Delhi",
  "seats": NumberInt(1),
  "total_fare": NumberInt(5000)
}
```

Q10. Update price by +500 for flights of airline "Indigo".

```
db.flightsCSF017.updateMany(
  { airline: "Indigo" },
  { $inc: { price: 500 } }
)
```

```
{
  "acknowledged": true,
  "insertedId": null,
  "matchedCount": 36.0,
  "modifiedCount": 36.0,
  "upsertedCount": 0.0
}
```

Q11. Delete bookings where seats are greater than 2.

```
db.bookingsCSF017.deleteMany({ seats: { $gt: 2 } })
```

```
1 {
2   "acknowledged" : true,
3   "insertedId" : null,
4   "matchedCount" : 38.0,
5   "modifiedCount" : 38.0,
6   "upsertedCount" : 0.0
7 }
8 {
9   "acknowledged" : true,
10  "deletedCount" : 18.0
11 }
12
```

Q12. Find total fare collected city-wise. (Aggregation)

```
db.bookingsCSF017.aggregate([
  {
    $group: {
      _id: "$city",
      totalFare: { $sum: "$total_fare" }
    }
  }
])
```

```
1 {
2   "_id" : "Bangalore",
3   "totalFare" : NumberInt(104500)
4 }
5 {
6   "_id" : "Kolkata",
7   "totalFare" : NumberInt(247000)
8 }
9 {
10  "_id" : "Delhi",
11  "totalFare" : NumberInt(95000)
12 }
13 {
14  "_id" : "Mumbai",
15  "totalFare" : NumberInt(258400)
16 }
17
```

Q13. Find average flight price per airline. (Aggregation)

```
db.flightsCSF017.aggregate([
  {
    $group: {
      _id: "$airline",
      avgPrice: { $avg: "$price" }
    }
  }
])
```

```
1 {
2   "_id" : "Vistara",
3   "avgPrice" : 7000.0
4 }
5 {
6   "_id" : "AirIndia",
7   "avgPrice" : 6500.0
8 }
9 {
10  "_id" : "Indigo",
11  "avgPrice" : 7107.142857142857
12 }
13 {
14  "_id" : "SpiceJet",
15  "avgPrice" : 4000.0
16 }
17 {
18  "_id" : "GoFirst",
19  "avgPrice" : 4800.0
20 }
21
```

Q14. Display airlines where average price is greater than 5500. (Aggregation + filter)

```
db.flightsCSF017.aggregate([
{
  $group: {
    _id: "$airline",
    avgPrice: { $avg: "$price" }
  }
}]
)
```

50 Documents 1 to 5 Query Code Explain Query JSON View

```
1 {
2   "_id" : "Vistara",
3   "avgPrice" : 7000.0
4 }
5 {
6   "_id" : "AirIndia",
7   "avgPrice" : 6500.0
8 }
9 {
10  "_id" : "Indigo",
11  "avgPrice" : 7107.142857142857
12 }
13 {
14  "_id" : "SpiceJet",
15  "avgPrice" : 4000.0
16 }
17 {
18  "_id" : "GoFirst",
19  "avgPrice" : 4800.0
20 }
21
```

Q15. Find maximum total_fare among bookings for each city. (*Aggregation*)

```
db.bookingsCSF017.aggregate([
{
  $group: {
    _id: "$city",
    maxFare: { $max: "$total_fare" }
  }
}]
)
```

Find Query (1) Shell Output (D) Aggregate Quer... Aggregate Quer... Aggregate Quer... Find Query (1) Shell Output (D) Aggregate Quer... Aggregate Quer... Aggregate Quer... 50 Documents 1 to 4 Query Code Explain Query JSON View

```
1 {
2   "_id" : "Kolkata",
3   "maxFare" : NumberInt(13000)
4 }
5 {
6   "_id" : "Delhi",
7   "maxFare" : NumberInt(5000)
8 }
9 {
10  "_id" : "Mumbai",
11  "maxFare" : NumberInt(9600)
12 }
13 {
14  "_id" : "Bangalore",
15  "maxFare" : NumberInt(5500)
16 }
17
```